The cardiac muscle

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objectives

- **Cardiac action potential**
- **Pacemaker potential**
- **AV delay - significance**
- **Cardiac contractility is increased under the influence of hormone Norepinephrine.**

Functions of the Heart

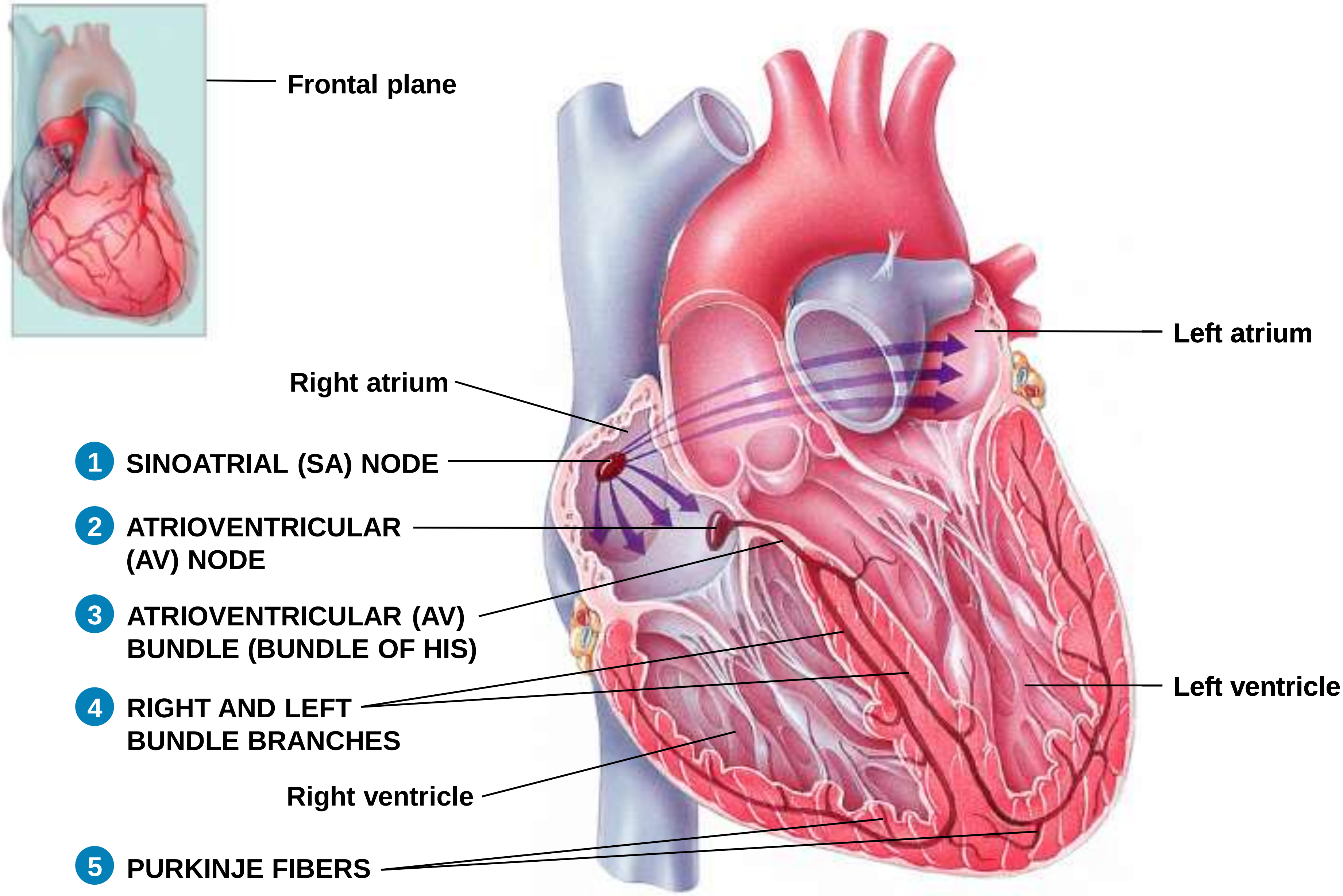
- Generating blood pressure
- Routing blood: separates pulmonary and systemic circulations
- Ensuring one-way blood flow: valves
- Regulating blood supply
  - Changes in contraction rate and force match blood delivery to changing metabolic needs
  - Exercise
  - Sleeping
  - Sitting

Autorhythmic Fibers

- Specialized cardiac muscle fibers
- Self-excitable
- Repeatedly generate action potentials that trigger heart contractions
- 2 important functions
  1. Act as pacemaker
  2. Form conduction system

Conduction system

1. Begins in sinoatrial (SA) node in right atrial wall
   - Propagates through atria via gap junctions
   - Atria contract
2. Reaches atrioventricular (AV) node in interatrial septum
3. Enters atrioventricular (AV) bundle (Bundle of His)
   - Only site where action potentials can conduct from atria to ventricles due to fibrous skeleton
4. Enters right and left bundle branches which extends through interventricular septum towards apex
5. Finally, large diameter Purkinje fibers conduct action potential to remainder of ventricular myocardium
   - Ventricles contract



Anterior view of frontal section

Conduction System

- SA node acts as natural pacemaker
  - Faster than other autorhythmic fibers
  - Initiates 100 times per second
- Nerve impulses from autonomic nervous system (ANS) and hormones modify timing and strength of each heartbeat
  - Do not establish fundamental (constant) rhythm

Action Potentials and Contraction

- Action potential initiated by SA node spreads out to excite “working” fibers called contractile fibers

1. Depolarization
2. Plateau
3. Repolarization

Action Potentials and Contraction

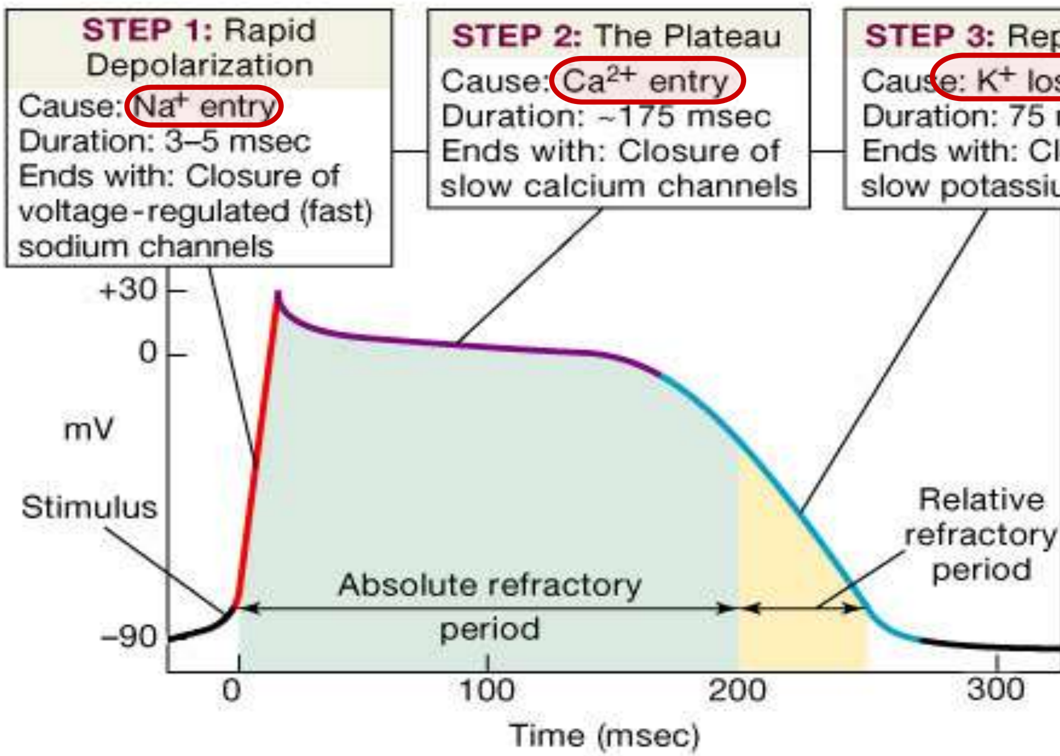
1. Depolarization – contractile fibers have stable resting membrane potential
   - Voltage-gated fast Na^+ channels open – Na^+ flows in
   - Then deactivate and Na^+ inflow decreases
2. Plateau – period of maintained depolarization
   - Due in part to opening of voltage-gated slow Ca^{2+} channels – Ca^{2+} moves from interstitial fluid into cytosol
   - Ultimately triggers contraction
   - Depolarization sustained due to voltage-gated K^+ channels balancing Ca^{2+} inflow with K^+ outflow

Action Potentials and Contraction

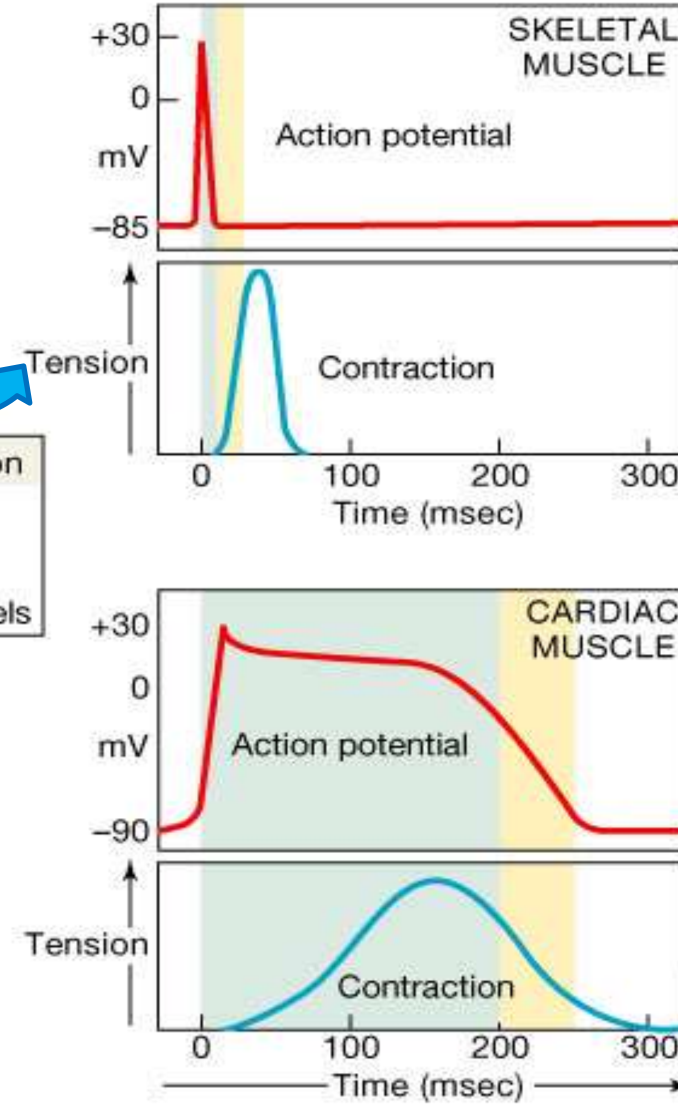
3. Repolarization – recovery of resting membrane potential
    - Resembles that in other excitable cells
    - Additional voltage-gated K^+ channels open
    - Outflow of K^+ restores negative resting membrane potential
    - Calcium channels closing

- Refractory period – time interval during which second contraction cannot be triggered
    - Lasts longer than contraction itself
    - Hence Tetanus (maintained contraction) cannot occur
        - Then, Blood flow would cease

TWITCH

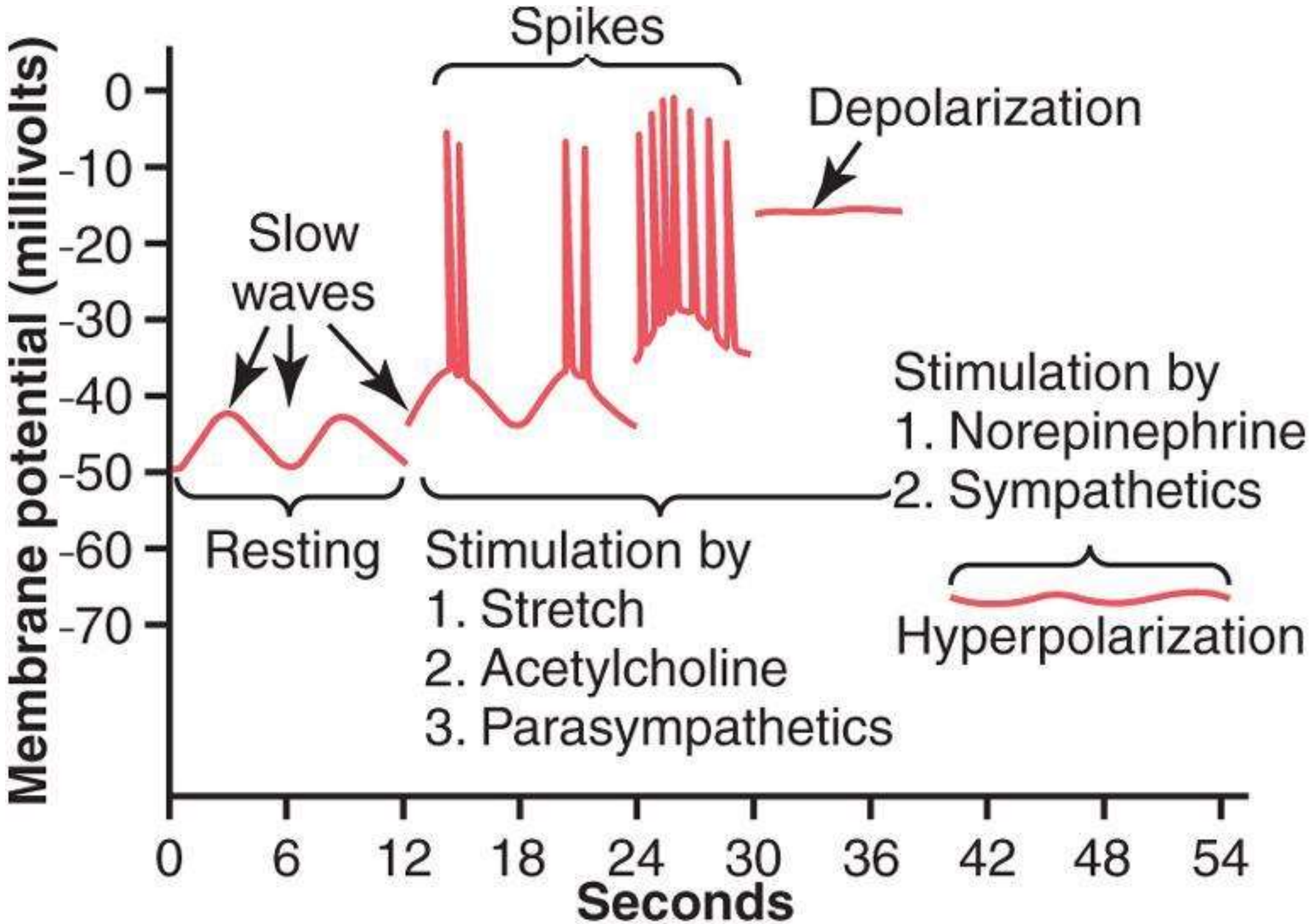


(a) Cardiac muscle



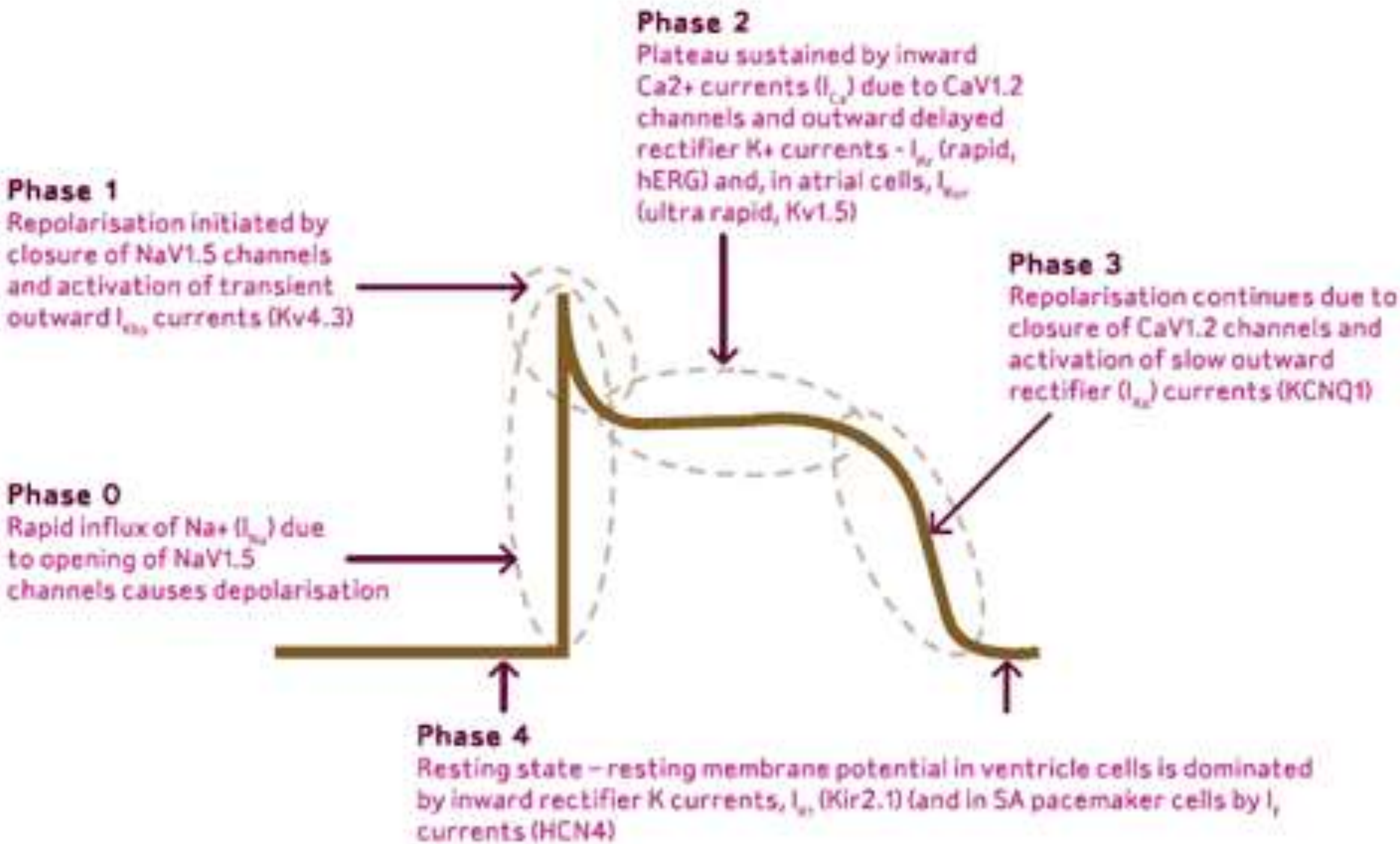
(b)

Smooth muscles action potentials in Gut

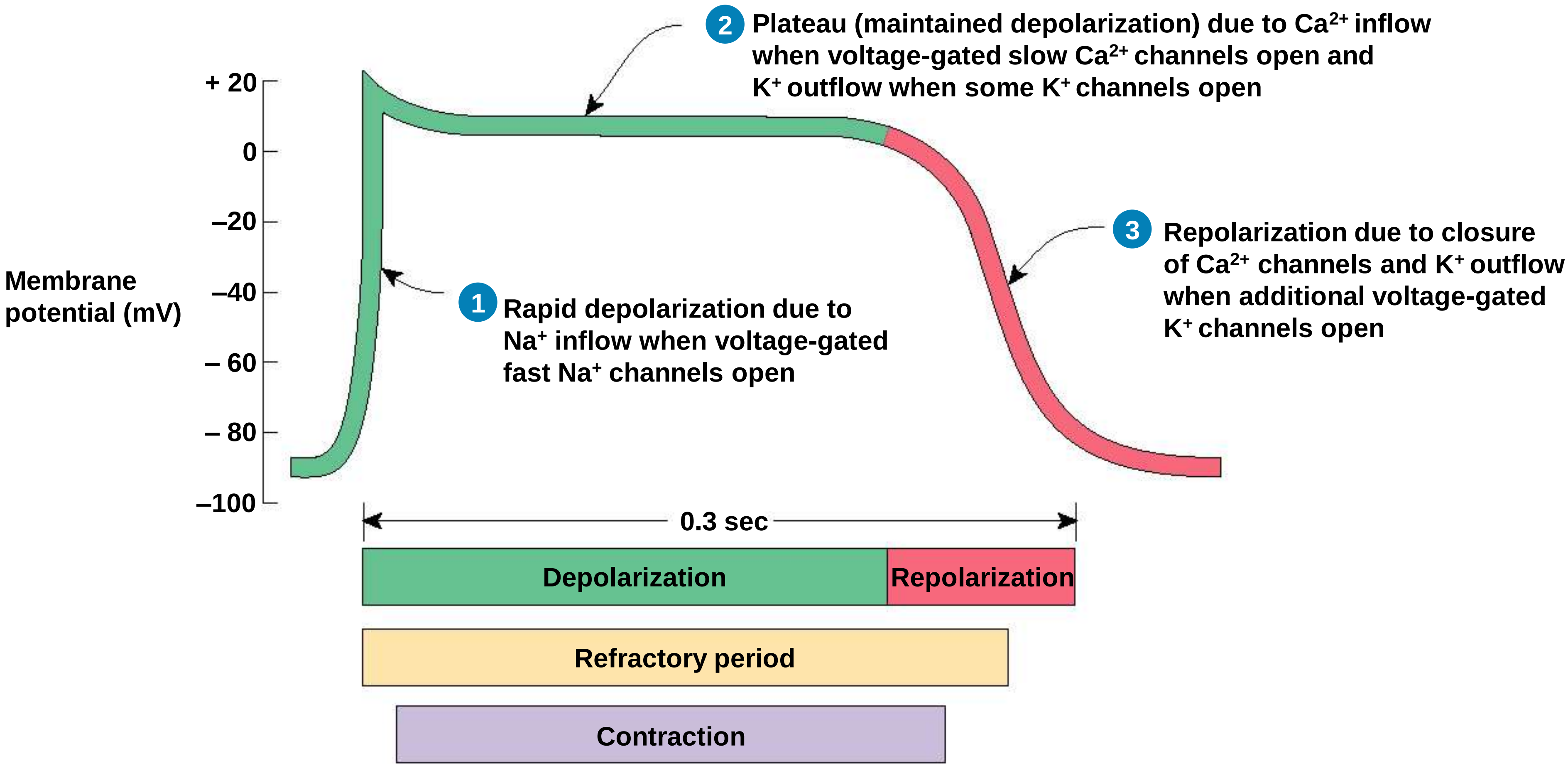


Guyton's textbook

cardiac



Phase 2
Plateau sustained by inward Ca2+ currents (I_{Ca}) due to CaV1.2 channels and outward delayed rectifier K+ currents - I_{Kr} (rapid, hERG) and, in atrial cells, I_{Kur} (ultra rapid, Kv1.5)
Phase 1
Repolarisation initiated by closure of NaV1.5 channels and activation of transient outward I_{Kto} currents (Kv4.3)
Phase 3
Repolarisation continues due to closure of CaV1.2 channels and activation of slow outward rectifier (I_{Ks}) currents (KCNQ1)
Phase 0
Rapid influx of Na+ (I_{Na}) due to opening of NaV1.5 channels causes depolarisation
Phase 4
Resting state – resting membrane potential in ventricle cells is dominated by inward rectifier K currents, I_{K1} (Kir2.1) (and in SA pacemaker cells by I_f currents (HCN4)



Refractory period

When a muscle fiber contracts, it temporarily cannot respond to another action potential

- Skeletal muscle has a refractory period of 5 milliseconds
- Cardiac muscle has a refractory period of 300 milliseconds- so no tetanus.

- because Na^+ channels are inactivated during this time, additional depolarizing stimuli do not lead to new action potentials.

- Cardiac muscle cannot be tetanised because of long refractory period

Mechanism of Sinus nodal rhythmicity

- A process that cause automatic rhythmical discharge and contraction.
- RMP of sinus nodal fiber (SAN) is -55 to -60mv.(unstable RMP)
- **The inherent leakiness of sinus nodal fibers to sodium and calcium ions cause their self excitation.**

Only in living cells

- Pacemaker potential or Pre-potential -

The slow rising RMP in between action potentials is k/a pacemaker potential/prepotential.

Observe this heart !!

Are both atria and ventricles beating at the same time ?

What makes the heart to contract atria first then the ventricles?

AV node

The AV node is located in the posterior wall of the right atrium immediately behind the tricuspid valve.

AV nodal Delay

- Due to the slow conduction in AV node there is a delay of **0.09** sec in the AV node, before the impulse reaches the penetrating portion of AV bundle.
- There is a delay of **0.04** sec in the penetrating portion of AV bundle.
- There is a total delay of **0.13** sec in the AV node and AV bundle system.
- There is a delay of **0.03** sec from the sinus node to the AV node, thus making **total delay of 0.16 sec** before the excitatory signal finally reaches the contracting muscle of ventricles.

Cause of Slow Conduction in the A-V Node

The cause of slow conduction is mainly due to diminished number of gap junctions between the successive cells in the conducting pathways.

As a result of which there is great resistance to conduction of excitatory ions from one conducting fiber to the next.

Significance of AV nodal delay

- The cardiac impulse does not travel from the atria to the ventricles too rapidly.
- This delay allows time for the atria to empty their blood into the ventricles before ventricular contraction begins. This increases the efficiency of the pumping action of the heart.
- It is primarily the AV node and it’s adjacent fibers that delay this transmission into the ventricles

AV Bundle or Bundle of His

- From the AV node arises a special conducting pathway called the bundle of His.
- Except for the very small part which penetrates through the AV fibrous tissue and has low conduction velocity, the bundle of His gives rise purkinje fibers which possess maximum conduction velocity in the heart.

- Purkinje fibers are very large fibers and they transmit action potentials at a velocity of 1.5 to 4.0 m/sec.
- The rapid transmission of action potentials through the Purkinje fibers is believed to be caused by a very high level of permeability of gap junctions at the intercalated discs between the successive cells of Purkinje fibers.
- The rapid conduction through the purkinje fibers ensures that different parts of ventricles are excited almost simultaneously; this greatly increases the efficiency of heart as a pump.

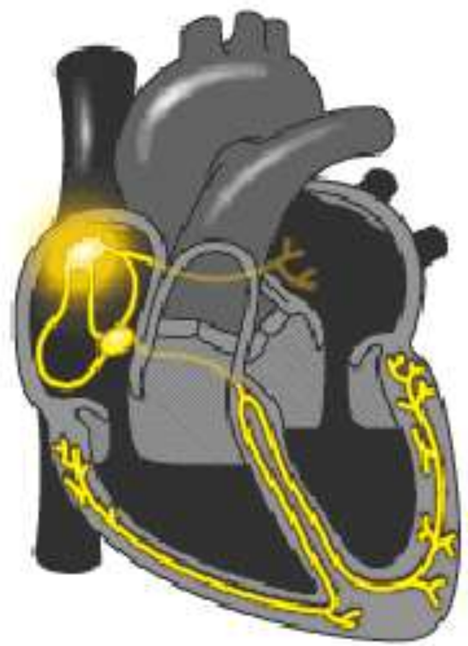
- Normally the <u>Bundle of His </u>is the only conducting mass <u>*between*</u> the atrial and ventricular musculature and it <u>transmits</u> the cardiac impulses from the AV node to the ventricles.

Right and Left Bundle Branches

- After penetrating the fibrous tissue between the atrial and ventricular muscle, the distal portion of the A-V bundle passes downward in the ventricular septum for 5 to 15 mm toward the apex of the heart.
- Then the bundle of His splits into two branches which are called right and left bundle branches that lie on the respective sides of the ventricular septum.

- Each branch spreads downward toward the apex of the ventricle, progressively dividing into smaller branches.
- These branches in turn course sidewise around each ventricular chamber and back toward the base of heart.
- The ends of Purkinje fibers penetrate about one third of the way into muscle mass and finally become continuous with cardiac muscle fibers.

- From the time the cardiac impulse enters the bundle branches until it reaches the terminations of Purkinje fibers , the total elapsed time averages only 0.03 sec.



- **Observe atria, then the ventricles**

One- way Conduction through AV bundle

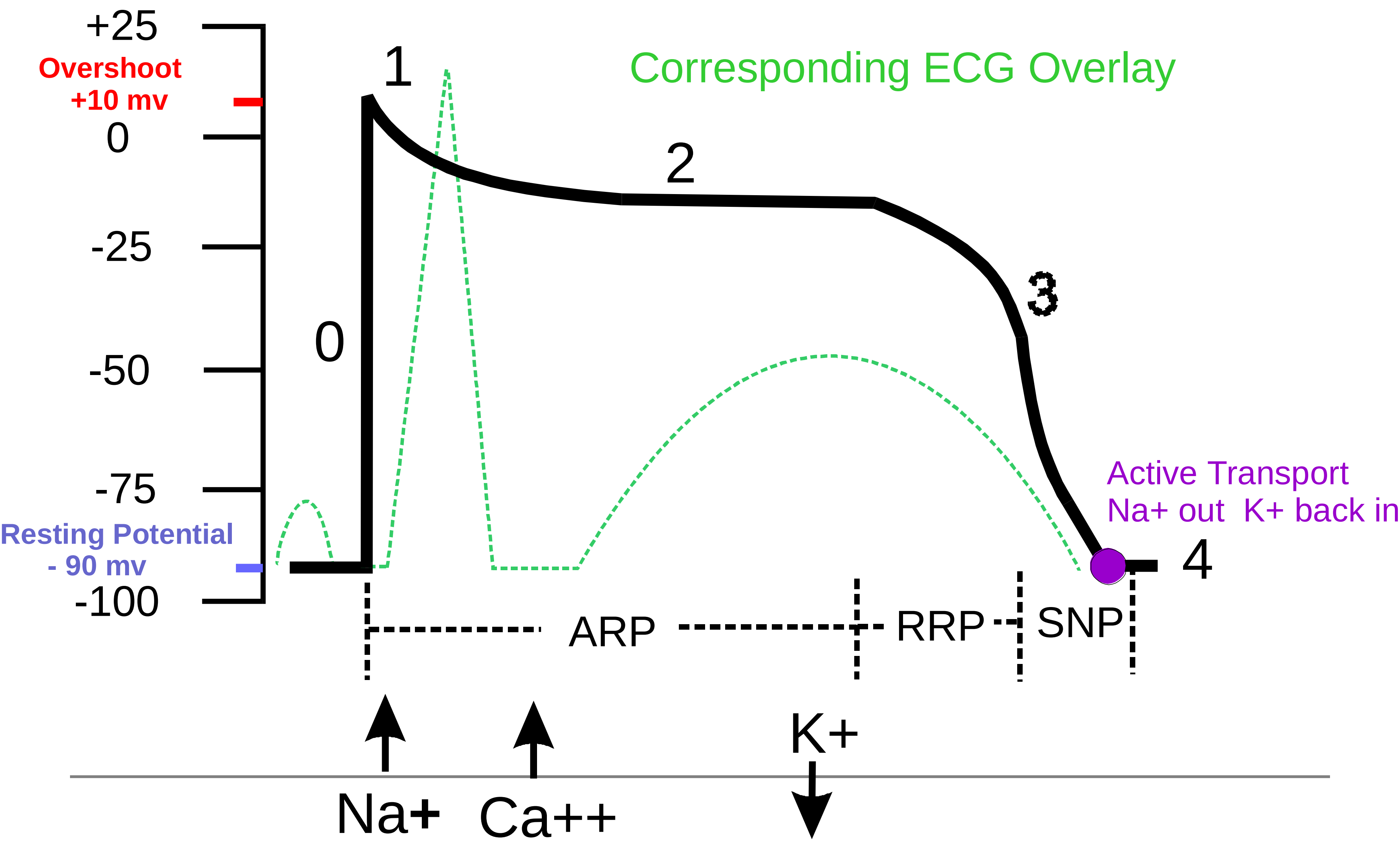
- A special characteristic of the A-V bundle is it's inability, except in the abnormal states , of action potentials to travel backward from the ventricles to the atria.
- This prevents re-entry of cardiac impulse by this route from the ventricles to the atria.
- The atrial muscle is separated from the ventricular muscle by a continuous fibrous barrier which acts as an insulator to prevent the passage of cardiac impulse between the atrial and ventricular muscle through any other route besides forward conduction through A-V bundle itself.

- **Sinoatrial (SA) node** normally generates the action potential, i.e. the electrical impulse that initiates contraction.
  - The SA node excites the right atrium (RA), travels through Bachmann's bundle to excite left atrium (LA).
  - The impulse travels through internodal pathways in RA to the **atrioventricular (AV) node.**

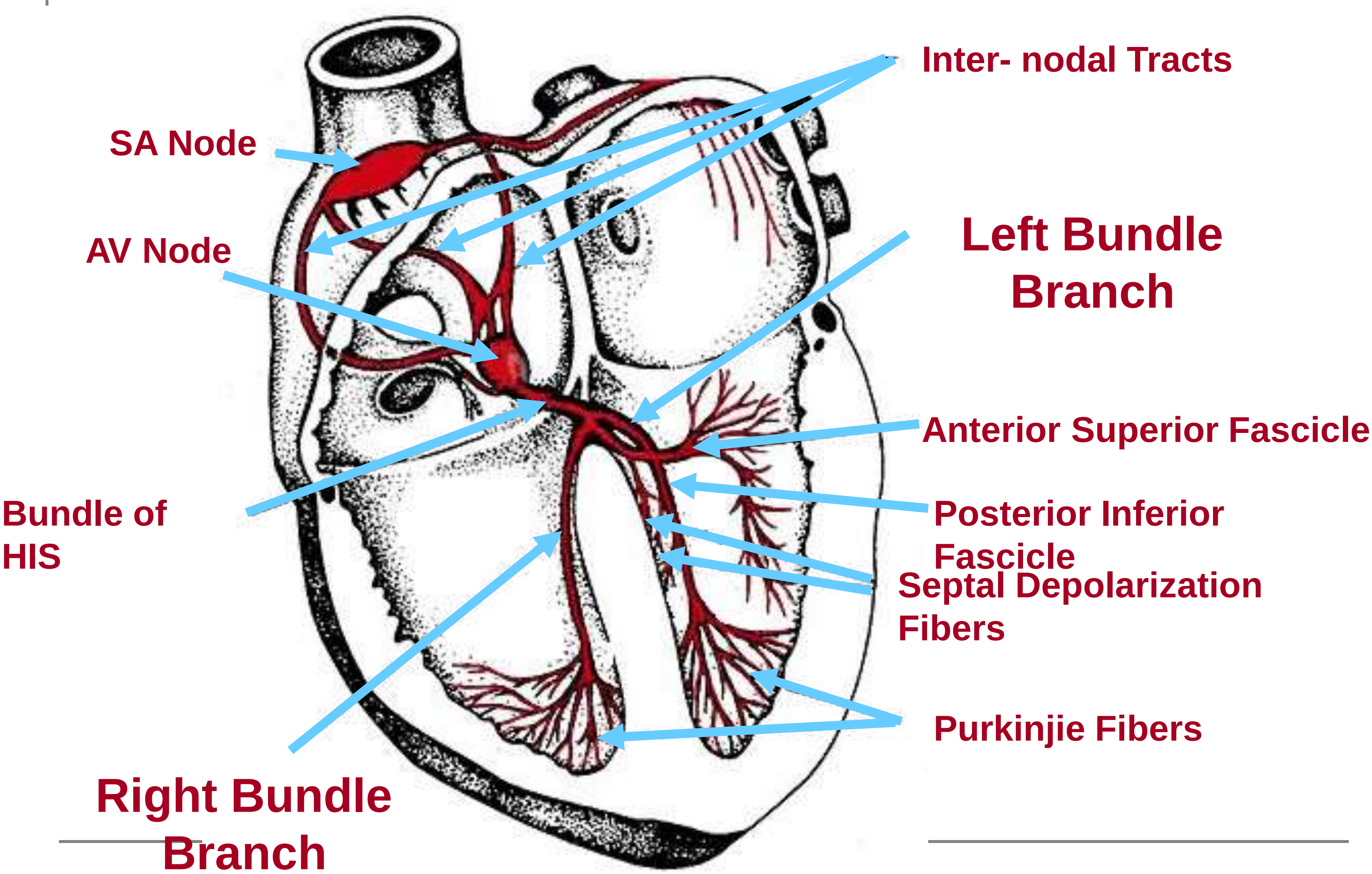
Questions ?

Does the action potential spreads from atrial muscles to the ventricles directly?

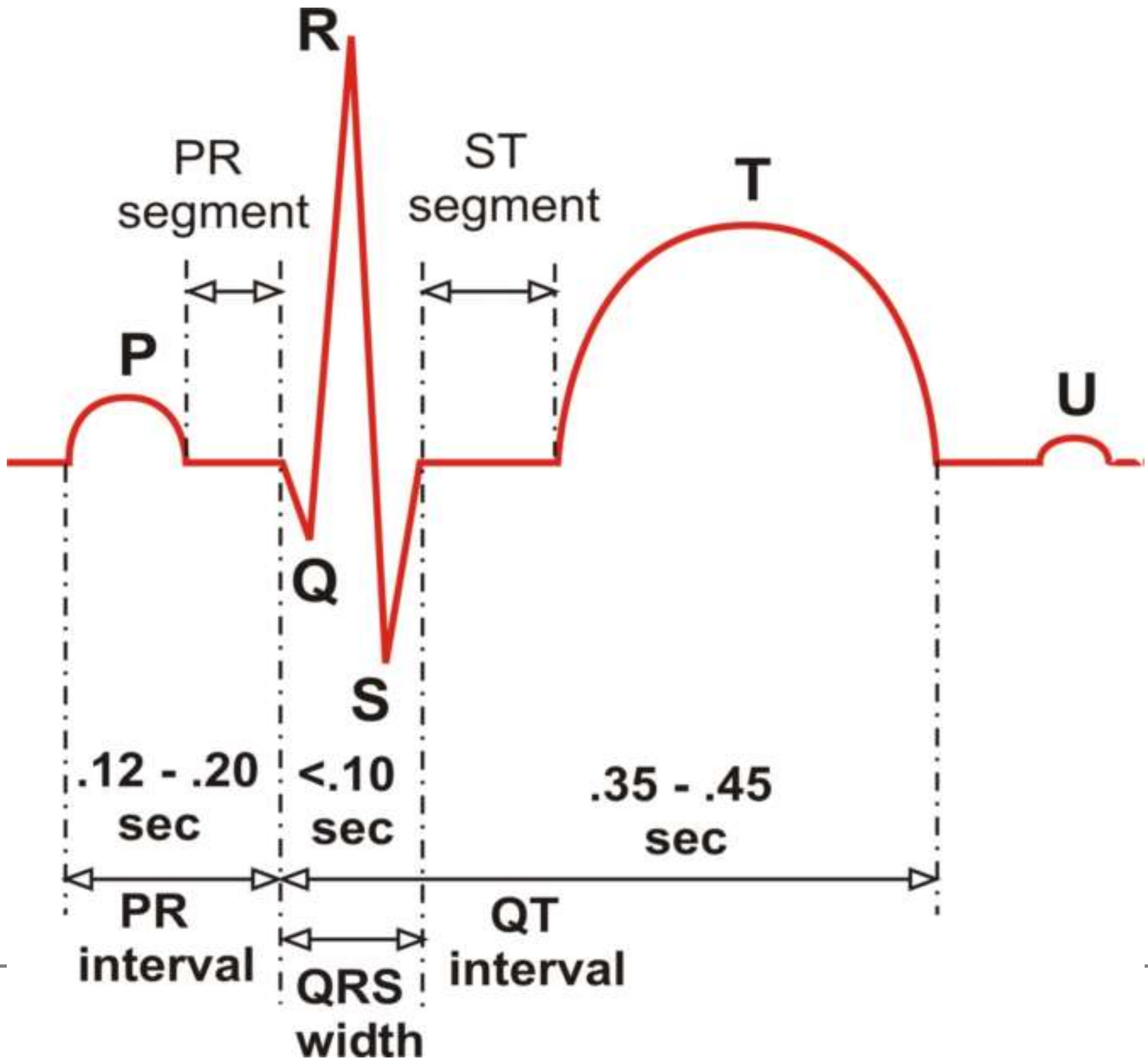
Myocardium Muscle Action Potential



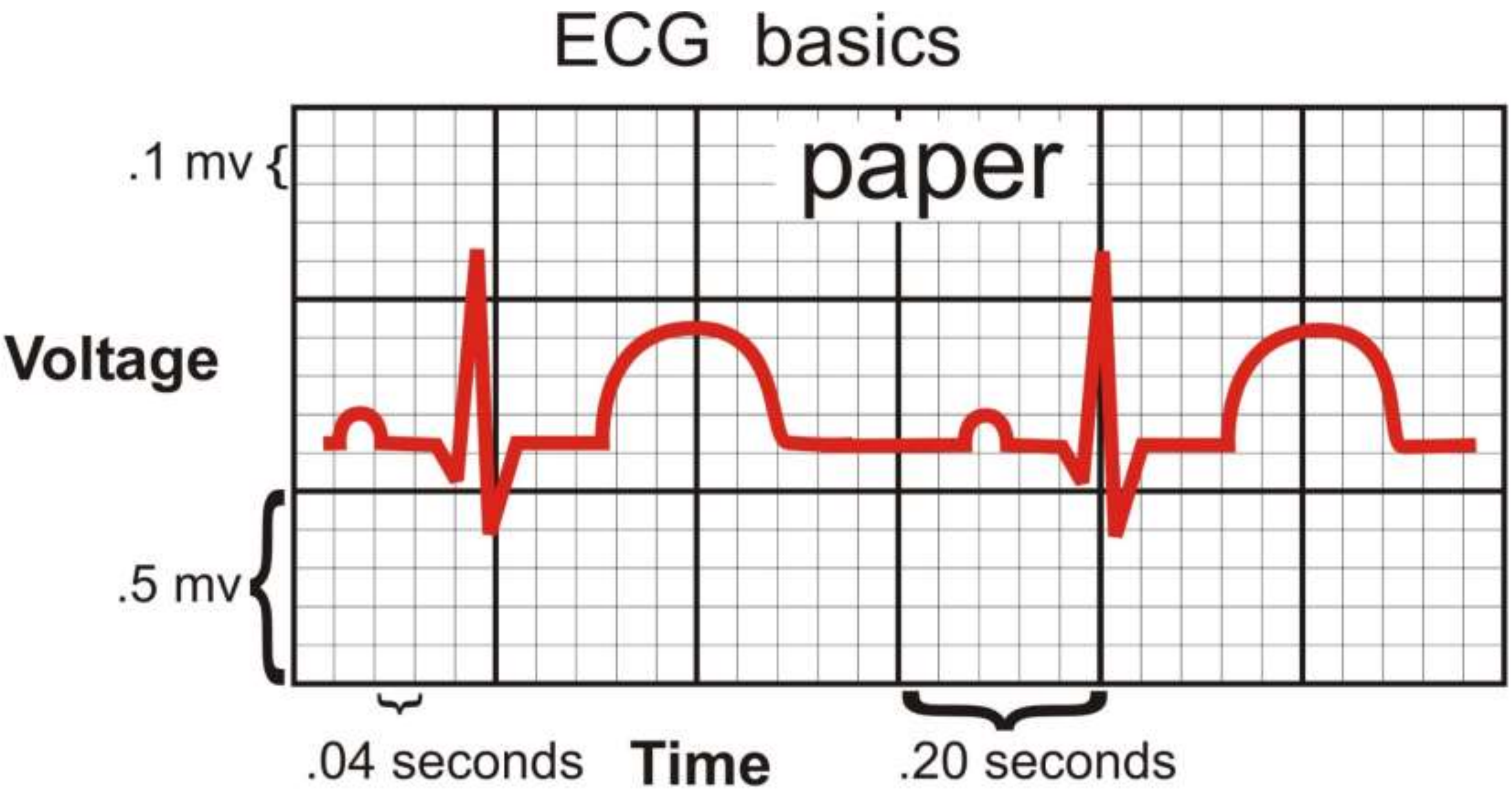
The Electrical System of the Heart



The ECG Complex with Interval and Segment Measurements



ECG Paper and related Heart Rate & Voltage Computations



Paper speed = 25mm / second

Heart Rate = number of R-waves in a 6 second strip divided by 10

= 1500 divided by the number of small boxes between consecutive R-waves

= large square estimation counts (300 - 150 - 100 - 75 - 60 - 50 - 43)

Memorize These 2

Questions ?